

Recovering Collective Structure in Congressional Voting: A Fused Latent Factor and Network Approach

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Draft: August 24, 2026

Abstract

Congressional voting reflects collective behavior structured by chamber-wide patterns such as partisanship and relational patterns among particular subsets of legislators, including factions and coalitions. Yet latent factor frameworks that dominate roll call analysis assume conditional independence across legislators and represent their ideal positions in a linear space. These restrictions can obscure relational patterns that do not align with chamber-wide latent dimensions. To model these behaviors, we adopt a fused latent factor and network model that captures chamber-wide patterns with the latent factors and attributes residual structure to the network. Applying this fused model to the 101st (1989–90) and 114th (2015–16) Senates, we first validate that the latent factors recover partisan voting patterns and identify two systematic failures of the latent factor: senators with moderate latent positions are assigned uncertain probabilities on party-line votes, and intra-party split votes exhibit systematic co-defection patterns. The network component closes the misprediction gap entirely for party-line votes but only partially for intra-party splits, capturing stable co-defection patterns while leaving more transient coalitions less predictable. Next, we apply a degree-corrected stochastic block model to the fitted network and recover interpretable voting blocs in both Senates. The 101st Senate features within-party factions that match historically documented coalitions and are driven by shared issue interests despite overlapping positions in the latent space. In the 114th Senate, the blocks take on an entirely different interpretation, uncovering two institutionally opposed coalitions: a cross-party governing bloc, including members of the Republican party leadership, and a Republican insurgent bloc, both driven by opposition of the other. Our findings demonstrate that roll call data alone contains recoverable signals of collective legislative behavior beyond partisanship that remain after accounting for chamber-wide partisan structure, and the organization of this collective behavior shifts from issue-based coalitions toward institutional conflict across decades of polarization.

Keywords: Roll Call Voting, Network Analysis, Latent Factor Models, Congressional Coalitions

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1 Introduction

Legislatures are collective institutions, and their outputs are often a result of coalitions, coordination, and extensive negotiation rather than individual action (Sinclair 2016). These collective patterns do not necessarily coincide with party. While some bills see exact party-line splits, the Civil Rights Act of 1964 was passed by a coalition that crossed party lines. Congress has also housed consequential within-party factions organized around shared regional, ideological, or policy interests, such as Southern Democrats and the Tea Party Caucus.

Capturing this variation in collective legislative behavior is therefore an important measurement problem. Two methods are commonly used in the political literature, but both have significant limitations. The first is the latent factor framework, which became the linchpin of roll call analysis with the introduction of NOMINATE (Poole and Rosenthal 1985) and subsequent models (Heckman and Snyder 1997; Clinton, Jackman, and Rivers 2004). Although these models have proven useful for uncovering partisan structure as a predominant feature of Congress, this framework imposes two restrictions on collective voting behavior. First, legislators’ voting choices are conditionally independent given the latent factors, which precludes any relational or coordinated behavior beyond what the latent factors capture. Second, latent factors represent collective patterns as chamber-wide continuous gradations along a common set of dimensions. Although such localized patterns may be accommodated through additional chamber-wide dimensions, this representation can be substantively misleading, because heterogeneity in collective behavior need not reflect high-dimensionality in individual political preferences.

Second, scholars emphasizing the relational aspects of legislative behavior have opted for network analysis of the Congress (Kirkland 2011; Kirkland and Gross 2014). Cosponsorship networks have been especially common for examining coalitions (Fowler 2006; Kim, Barnett, and Kwon 2010; Battaglini, Sciabolazza, and Patacchini 2020), and other relational sources such as shared committee memberships (Porter et al. 2005) and geographical contiguity (Kim, Barnett, and Kwon 2010) have been studied as proxies of social structure. These network approaches have two limitations. First, disentangling relational effects from the influence of partisanship is methodologically difficult. Second, connecting

observed networks to tangible political outcomes such as roll call votes requires reconciling exogenous datasets.

We start from the premise that collective roll call behavior can operate at two different levels: chamber-wide patterns such as partisanship and relational patterns among particular subsets of legislators. We therefore combine both approaches for roll call analysis using a fused latent factor and network model. Following the congressional literature’s emphasis on partisanship as the predominant mechanism, we capture chamber-wide patterns with latent factors and attribute residual structure to the network, allowing separate representations of chamber-wide and relational patterns. We fit this model using optimization techniques on the 101st and 114th Senate roll call data. We validate this decomposition empirically, showing that the fitted latent factors recover partisan voting patterns in both Senates.

This paper focuses on two main objectives. First, we examine whether systematic collective voting patterns remain after accounting for the chamber-wide structure captured by latent factors. We identify two classes of systematic failures of the latent framework associated with collective behavior: moderate legislators on party-line bills and intra-party split bills. The fitted network entirely closes the prediction gap for the former and captures stable co-defection patterns underlying the latter, while leaving more transient coalitions less predictable.

Second, we offer a substantive analysis of collective legislative behavior beyond partisanship, recovered from roll call data alone. Applying a degree-corrected stochastic block model to the fitted network, we identify interpretable voting blocs in the 101st (1989–90) Senate, differentiated by distinct issue interests despite their overlapping positions in the latent space. Extending this analysis to the 114th (2015–16) Senate after decades of partisan polarization, we find that the block structure reveals institutional opposition: the network distinguishes a cross-party governing bloc, including members of the Republican party leadership, from a Republican insurgent bloc. Together, these analyses show that the organization of collective behavior beyond partisanship shifts from substantially issue-based coalitions in the 101st Senate toward institutional conflict over governing in the 114th. They also demonstrate that relational patterns of collective legislative behavior can be recovered directly from roll call data without exogenous

measures of legislative relationships.

The rest of the paper is organized as follows. In section 2, we introduce a fused latent and network model and the computational method for its estimation. Section 3 presents the empirical results. We validate the fitted model, identify the systematic failures of the latent factor model, and analyze the fitted network to recover collective behavior. We also compare the 101st and 114th Senates to observe changes in coalition structure over time. Section 4 concludes.

2 A Fused Latent Factor and Social Network Model

To simultaneously capture the dominant partisan patterns of roll call voting and secondary coalitional efforts, we adopt a fused latent factor and social network model developed by Chen et al. (2016). In this model, votes are primarily attributed to latent factors, while any remaining variation is cast to a network. The latent factors represent the chamber-wide voting patterns, while the network explains deviations from this pattern as a relational structure inferred from legislators’ relatively atypical votes.

2.1 The M2PL-Ising Model

The multidimensional two-parameter logistic (M2PL)-Ising model, developed by Chen et al. (2016), brings together two models with historically different usage. The first is a common item response theory model for measuring latent traits given binary responses. Given a two-year term of congress with J legislators and N bills, with the j th legislator’s vote for the n th bill denoted by $Y_{jn} \in \{0, 1\}$, the M2PL model defines the probability of a “Yea” vote ($Y_{jn} = 1$) as a logistic distribution:

$$P(Y_{jn} = y_{jn} | \boldsymbol{\theta}_n, \mathbf{a}_j, d_j) = \frac{\exp\left\{(\mathbf{a}_j^\top \boldsymbol{\theta}_n + d_j)y_{jn}\right\}}{1 + \exp\left\{(\mathbf{a}_j^\top \boldsymbol{\theta}_n + d_j)y_{jn}\right\}} \propto \exp\left\{(\mathbf{a}_j^\top \boldsymbol{\theta}_n + d_j)y_{jn}\right\}. \quad (1)$$

Here, $\boldsymbol{\theta}_n \in \mathbb{R}^K$ denotes the $K > 1$ dimensional latent factors of the bill n , $\mathbf{a}_j \in \mathbb{R}^K$ the factor loadings or the discrimination parameter of the j th legislator, and d_j the difficulty parameter of the j th legislator. Proceeding with equation (1), a local independence assumption is typically made to obtain the joint

conditional distribution, with $A_{J \times K} = (\mathbf{a}_1, \mathbf{a}_2, \dots, \mathbf{a}_J)^\top$ and $\mathbf{d} = (d_1, d_2, \dots, d_J)^\top$:

$$P(\mathbf{Y}_n = \mathbf{y}_n | \boldsymbol{\theta}_n) \propto \exp\left\{\boldsymbol{\theta}_n^\top A^\top \mathbf{y}_n + \mathbf{d}^\top \mathbf{y}_n\right\}.$$

The M2PL model with $K = 2$ is analogous to the two-dimensional latent factor model of NOMINATE, with the key distinction being that NOMINATE uses a probit link to model $P(Y_{jn} = y_{jn})$, whereas M2PL uses a logit link (Clinton, Jackman, and Rivers 2004).

The second component is the Ising model, an exponential family graphical model originating from physics, which has been increasingly applied to social network settings to capture the conditional graphical dependence structure among a set of random variables (Stauffer 2008). Given binary random variables, each variable is associated with a vertex, and the undirected graphical structure among variables is mapped on a symmetric matrix $S = (s_{ij})_{J \times J}$ where s_{ij} —which can take positive, negative, and 0 values—indicates the “strength” of the edge between two variables. In the context of roll call votes, a larger s_{ij} indicates a higher probability that the i th and j th legislators will simultaneously vote “Yea”. The Ising model is defined as

$$P(\mathbf{Y}_n = \mathbf{y}_n) = \frac{1}{z(S)} \exp\left\{\frac{1}{2} \mathbf{y}_n^\top S \mathbf{y}_n\right\},$$

where $z(S)$ is the normalizing constant.

Taking $L = AA^\top$ with factor loading matrix A of the M2PL model and graph S of the Ising model, the M2PL-Ising model is a fusion of the two models, with the joint distribution given by

$$p(\mathbf{y}_n | L, S) = \frac{1}{z(L, S)} \exp\left\{\frac{1}{2} \mathbf{y}_n^\top (L + S) \mathbf{y}_n\right\}, \quad (2)$$

where $z(L, S)$ is again the normalizing constant. L captures the dependencies among the votes that are driven by the latent structure, which we interpret as ideology and partisanship, while S represents the network dependencies, which we interpret as social effects.

2.2 Estimation

The normalizing constant $z(L, S)$ from the model in equation (2) is computationally intractable. Therefore, to estimate L and S , Chen et al. (2016) propose a pseudolikelihood estimator. For j th legislator and n th bill, the pseudolikelihood is defined as

$$\begin{aligned}\mathcal{L}_{jn}(L, S; \mathbf{y}_n) &\equiv P(Y_{jn} = 1 | \mathbf{Y}_{-j,n} = \mathbf{y}_{-j,n}, L, S) \\ &= \frac{\exp\left\{\frac{l_{jj} + s_{jj}}{2} + \sum_{i \neq j} (l_{ij} + s_{ij}) y_{in}\right\}}{1 + \exp\left\{\frac{l_{jj} + s_{jj}}{2} + \sum_{i \neq j} (l_{ij} + s_{ij}) y_{in}\right\}}, \\ \mathcal{L}(L, S) &= \prod_{n=1}^N \prod_{j=1}^J \mathcal{L}_{jn}(L, S; \mathbf{y}_n).\end{aligned}\tag{3}$$

The pseudolikelihood in equation (3) depends on parameters L and S only through $L + S$. Therefore, additional assumptions are necessary to identify L and S from the data. To address this, Chen et al. assume that most of the dependence in Y is captured by the latent factors, and the remaining dependence is attributed to the graphical structure. This leads to the specification of L as a low-rank matrix and S as a symmetric sparse matrix. This specification has important implications for roll call analysis as it allows a decomposition of latent and network effects with the latent effect taking precedence.

We estimate L and S by minimizing the pseudolikelihood with two regularization terms: the nuclear norm of L to induce the low rank of L , and an L_1 regularization term of S to induce element-wise sparsity (Wright and Ma 2022), with ρ and γ as tuning parameters. We additionally assume that L is positive semi-definite and S is symmetric.

$$(\hat{L}, \hat{S}) = \arg \min_{L, S} \left\{ -\frac{1}{N} \log\{\mathcal{L}(L, S)\} + \rho \|L\|_* + \gamma \sum_{i \neq j} |s_{ij}| \right\}$$

To fit the pseudolikelihood estimator, we follow Chen et al. and implement the alternating direction method of multipliers (ADMM) algorithm (Boyd et al. 2011), a popular method for convex optimization.

For the objective above, this takes following form:

$$\min_{\mathbf{x}, \mathbf{z}} f(\mathbf{x}) + g(\mathbf{z}) \quad \text{s.t. } \mathbf{x} = \mathbf{z},$$

where $\mathbf{x} = (M, L, S)$,

$$\mathbf{z} = (\tilde{M}, \tilde{L}, \tilde{S}),$$

$$f(\mathbf{x}) = \begin{cases} h(M) + \rho \|L\|_* + \gamma \sum_{i! = j} |s_{ij}|, & \text{if } L \succeq, S^\top = S \\ \infty, & \text{otherwise} \end{cases}$$

$$g(\mathbf{z}) = \begin{cases} 0, & \text{if } M^\top = M, M = L + S \\ \infty, & \text{otherwise} \end{cases}$$

We execute the following updates until all primal and dual residuals meet the convergence criteria, with step size λ :

$$\mathbf{x}^{k+1} = \arg \min_{\mathbf{x}} \left\{ f(\mathbf{x}) + \frac{1}{2\lambda} \left\| \mathbf{x} - (\mathbf{z}^k - \mathbf{u}^k) \right\|_2^2 \right\},$$

$$\mathbf{z}^{k+1} = \arg \min_{\mathbf{z}} \left\{ g(\mathbf{z}) + \frac{1}{2\lambda} \left\| \mathbf{z} - (\mathbf{z}^k - \mathbf{u}^k) \right\|_2^2 \right\},$$

$$\mathbf{u}^{k+1} = \mathbf{u}^k + \mathbf{x}^{k+1} - \mathbf{z}^{k+1}.$$

This algorithm was implemented in Python, with the code available on GitHub. We fit multiple models $\mathcal{M}^{\rho, \gamma}$ over ranges of the tuning parameters ρ, γ and select the model that minimizes the Bayesian Information Criterion (BIC),

$$\text{BIC}(\mathcal{M}^{\rho, \gamma}) = -2 \log \mathcal{L}(\hat{L}^{\rho, \gamma}, \hat{S}^{\rho, \gamma}) + |\mathcal{M}^{\rho, \gamma}| \log N.$$

3 Results

Using the M2PL-Ising model, we analyze roll call votes in the 101st and 114th Senates. We choose the Senate over the House for its smaller size, less hierarchical structure, and its unique structural features of

unanimous consent agreements and filibusters (Sinclair 2016; Wawro and Schickler 2007). These features lead to more relation-based legislating that relies on informal negotiations and cooperation beyond formal rules and procedures that potentially serves as a rich source of relational structure for the fitted network.

We first validate the fitted $\hat{L}$ and $\hat{S}$ with the 101st Senate. Then, we identify the systematic failures of the latent factor model and analyze voting blocs in the fitted network. Lastly, we compare the analyses from the 101st and 114th Senates to characterize changes in congressional collective behavior and its interpretations.

3.1 Validation

Model	S101		S114	
	Accuracy	APRE	Accuracy	APRE
Party baseline	83.6%	0.348	90.5%	0.682
NOMINATE 1D (senator-CV)	85.1%	0.408	91.9%	0.730
NOMINATE 2D (senator-CV)	86.1%	0.448	91.8%	0.728
Latent $\hat{L}$ only	84.3%	0.373	91.6%	0.719
Full $\hat{L} + \hat{S}$	89.0%	0.560	93.8%	0.794

Table 1: Predictive accuracy and aggregate proportional reduction in error (APRE) for the 101st and 114th Senates. APRE is computed relative to a naive baseline of predicting each bill’s majority side.

Table 1 displays predictive accuracy across model specifications for both the 101st and 114th Senates. Our latent model alone produces comparable accuracy relative to NOMINATE, while the full latent and graphical model raises accuracy by 4.7 and 2.3 percentage points. In this section, we examine what known features of roll call behavior our estimated $\hat{L}$ and $\hat{S}$ capture.

3.1.1 Estimated Latent Factors

We first validate the estimated latent factors through the 101st Senate. We restricted our model selection to those with latent dimensions of 2 or higher for consistency with other latent factor models that typically use a two-dimensional factor space. Our final choice is the model with $\text{rank}(\hat{L}) = 2$ and the lowest BIC.

In our application, θ indicates the latent factors of the bills, in contrast to existing latent factor

applications where it typically corresponds to the ideal points of the legislators. While we do not directly estimate the ideal points of senators, we can use A , the factor loadings of senators, as a proxy to examine the ideal points of senators. Two features of A make it a useful proxy for senator latent factors: the sign of the slope, which indicates whether the senator prefers bills aligned with Democratic or Republican preferences, and the magnitude of the slope, which reflects the degree of partisan division of the senator’s voting patterns.

Following Chen et al., we first extract the factor loadings $\hat{A}$ from $\hat{L} = \hat{A}\hat{A}^\top$. Factor loadings are not unique and dependent on the method for extraction. We first obtain $\hat{A}_{\text{var}}$ using varimax rotation and also estimate the latent factor of bills, $\hat{\theta}_{\text{var}} = \hat{A}_{\text{var}}^\top Y$. We find in Table 2 that the first dimension of the estimated latent factors ($\hat{\theta}_{\text{var},1}$) is highly correlated (-0.984) with the proportions of “Yea” votes among Democrats for each bill (p_D). Similarly, the second dimension of factors ($\hat{\theta}_{\text{var},2}$) is highly correlated (0.988) with the proportion of “Yea” votes among Republicans for each bill (p_R). These high correlations demonstrate that the estimated factors successfully capture patterns of partisan voting.

Comparing the two dimensions of the discrimination parameters ($\hat{A}_{\text{var},1}$ and $\hat{A}_{\text{var},2}$) with two-dimensional DW-NOMINATE scores in Table 3 reveals that the discrimination parameters are heavily correlated with the first dimension of nominate scores. These correlations are consistent with Table 2, showing that the partisan dynamics famously captured by the first dimension of NOMINATE are also captured by $\hat{L}$,

Our latent factors, however, are not identical to those of NOMINATE. Notably, Table 3 also reveals that the second dimension of NOMINATE scores are not captured by our latent factors. To better understand what our latent factors do capture, we obtain the principal axes $\hat{A}_{\text{pr}}$ of $\hat{L}$, an alternate basis obtained through eigen-decomposition for the same rank-2 subspace, and estimate a second set of bill latent factors $\hat{\theta}_{\text{pr}}$. Table 4 explores the correlations between these bill latent factors and each of party split (yea-share difference between the two parties) and proportion of yea share per bill. We see medium correlation between first dimension of latent factors and both party split and yea share, which indicates that this dimension is capturing partisan dynamics equivalent to that of the first dimension of NOMINATE scores.

	p_D	p_R
$\hat{\theta}_{\text{var},1}$	-0.984	-0.047
$\hat{\theta}_{\text{var},2}$	+0.128	+0.988

Table 2: Correlation between the estimated latent bill factors ($\hat{\theta}_{\text{var},1}, \hat{\theta}_{\text{var},2}$) and the proportion of Democratic and Republican yea votes per bill (p_D, p_R) for the 101st Senate.

	$\hat{A}_{\text{var},1}$	$\hat{A}_{\text{var},2}$
NOM dim 1	+0.927	+0.889
NOM dim 2	-0.002	-0.051

Table 3: Correlation between DW-NOMINATE scores and the estimated factor loadings ($\hat{A}_{\text{var},1}, \hat{A}_{\text{var},2}$) for the 101st Senate.

	$\rho(\cdot, \text{party split})$	$\rho(\cdot, \text{yea share})$
$\hat{\theta}_{\text{pr},1}$	+0.424	-0.550
$\hat{\theta}_{\text{pr},2}$	-0.157	+0.825

Table 4: Correlation between the principal-axis latent factors ($\hat{\theta}_{\text{pr},1}, \hat{\theta}_{\text{pr},2}$) and two bill-level quantities for the 101st Senate: the absolute Democratic–Republican yea-share difference (party split, $|p_D - p_R|$) and the chamber yea share.

On the other hand, the second dimension of latent factors shows high correlation with yea shares. This illustrates one of the most distinctive differences between NOMINATE and our latent factors: for the former, consensus is absorbed by the bill-level cutpoint parameter dividing yea and nay voters; for the latter, consensus is instead captured by the senator-side factor loadings A . We revisit this feature in a future section, where it illustrates a crucial failure of latent factor models in capturing group behavior.

3.1.2 Estimated Network

The fitted graphical matrix $\hat{S}$ for the selected model has sparsity 0.114 with 553 edges, 492 positive and 61 negative. To validate $\hat{S}$, we examine whether it captures relationships that are at least partially exogenous to the patterns captured in the latent factor model. We regress $\hat{S}$ onto five observable networks using dyadic regression (Fafchamps and Gubert 2007) suitable for networks, which accounts for the possible

correlations between the edges associated with each node.

Cosponsorship-based networks have been popular for studying relational legislative behavior in the Congress (Fowler 2006; Kim, Barnett, and Kwon 2010; Battaglini, Sciabolazza, and Patacchini 2020). In his cosponsorship network, Fowler (2006) constructs the network edges based on direct sponsor-cosponsor relationships, and indirect connections between cosponsors are not reflected on the edge weights. However, given that there can be only one direct sponsor of the bill even though many bills reflect collaboration, we also include indirect cosponsor-cosponsor relationships in our cosponsorship network. Therefore, the edge weight between two senators corresponds to the number of all co-signed bills, either as sponsors or cosponsors.

Geographical coalitions that involve cooperation across party lines have been a longstanding feature in the Congress, with polarization between these coalitions intensifying in recent decades (Abramowitz 2012). In this context, representing neighboring constituencies is often considered a firm basis for political cooperation between legislators (Arnold, Deen, and Patterson 2000; Fowler 2006; Kim, Barnett, and Kwon 2010). We construct a geographical neighbor network with edge weights of 1 between senators who live in neighboring states, 0 otherwise.

Shared committee membership edges are coded as the number of common committee and subcommittees between two senators. Shared religion and veteran status networks assign binary edges between senators sharing the same religious affiliation or veteran status, respectively.

Predictor	$\hat{\beta}$	Dyadic SE	p -value
Intercept	-0.0905	0.0146	1.4×10^{-8} ***
Committee	+0.0001	0.0011	0.90
Cosponsorship	+0.0006	0.0001	3.4×10^{-14} ***
Geographic neighbor	+0.0084	0.0014	2.1×10^{-8} ***
Religion	+0.0072	0.0055	0.19
Veteran	-0.0061	0.0029	0.038 *
$R^2 = 0.047$, * $p < 0.05$, *** $p < 0.001$			

Table 5: Joint regression of $\hat{S}$ edge weights on observable covariate networks, with dyadic-robust standard errors (Fafchamps and Gubert 2007). There are $n = 4,851$ dyads, and dyadic degree of freedom is 98. The veteran coefficient, while nominally significant, loses significance after Bonferroni correction.

Table 5 reports the joint regression of $\hat{S}$ on all five networks. Cosponsorship and geographic neighbor networks significantly predict $\hat{S}$, while the other three networks do not contribute any additional explanatory power. The R^2 of 0.047 appears modest in absolute terms but is consistent with $\hat{S}$'s effective rank of 85.5, indicating a high-dimensional dependence structure encoded in $\hat{S}$ which far surpasses that of any single observable network (1.4–9.7).

3.2 Failures of the Latent Factor Model

The latent factor framework models congressional voting as the outcome of individual senator positions on a low-dimensional spatial model, assuming conditional independence across votes given these latent positions. This assumption precludes relational or coordinated behavior beyond the chamber-wide patterns captured by latent factors and produces two distinct classes of latent-prediction failure associated with collective behavior: on the senator level, positional moderates are systematically assigned uncertain probabilities, especially for party-line votes; on the bill level, intra-party split bills for which group divisions are poorly represented by the latent factors.

For analysis, we define party-line bills as those where two parties differ by at least 50 percentage points and each party has less than 15% minority. Positionally moderate senators are chosen as the bottom third of positional extremity, measured as each senator's distance from the between-party midpoint of the factor loading dimension of $\hat{A}$ most aligned with party membership. The prediction error is measured by the mean absolute error (MAE) for each senator, the average absolute difference between the senator's observed vote and their model-predicted probability of a yea vote, $\hat{p}_{jn} = \mathbb{E}[Y_{jn} \mid \mathbf{y}_{-jn}, \hat{L}] = \sigma(\hat{\alpha}_j + \sum_{k \neq j} \hat{L}_{jk} y_{kn})$.

3.2.1 Moderate Positions and Party-line Votes

In a latent factor framework, senators with moderate latent positions are systematically assigned mid-range probabilities. This connection inherently conflates moderateness of position with an uncertainty of a yea-vote. This ambiguity is especially notable for bills that produce clear party-line divides, resulting in a high prediction error for medium-position senators.

Model	Party-line MAE	Other MAE	Gap	p
Latent only	0.414	0.319	+0.096	< 0.001
Full $\hat{L} + \hat{S}$	0.237	0.212	+0.025	0.168

Table 6: Prediction error for positionally moderate senators on party-line versus all other votes. 101st Senate.

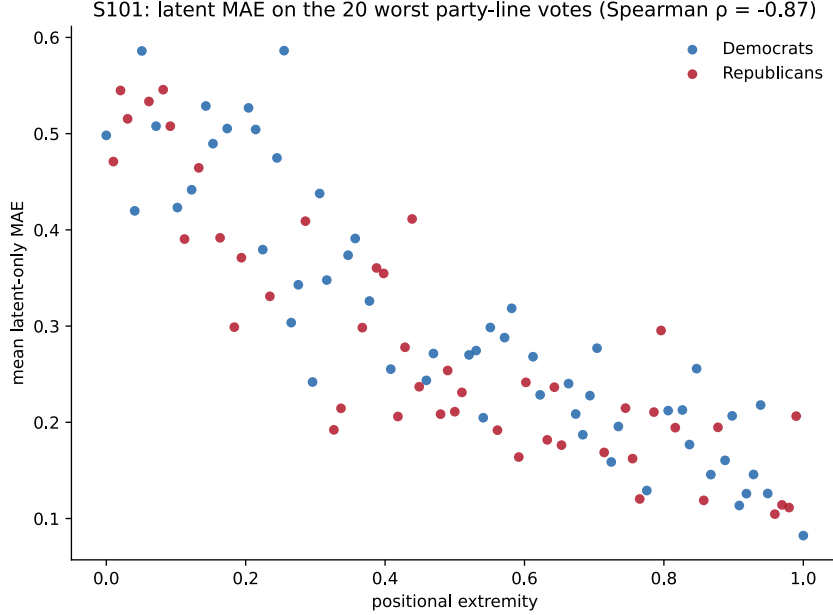


Figure 1: Senator positional extremity versus MAE on party-line votes, 101st Senate. Senators in the upper-left quadrant corresponding to low extremity and high prediction error are the positional moderates. Spearman $\rho = -0.87$ ($p < 0.001$).

We first inquire through a hypothesis test whether the latent model is systematically worse at predicting party-line bills than all other bills for positionally moderate senators. The first row of Table 6 reports the result of a 10,000-draw permutation test, which confirms that there is a significant difference in the prediction error for positionally moderate senators between party-line versus all other bills. We further illustrate this error by comparing senators’ positional extremity with prediction error across 20 worst latent-predicted party-line votes. Figure 1 displays a strong negative correlation ($\rho = -0.87, p < 0.001$), indicating that positionally moderate senators with lower extremities are associated with a systematically high MAE.

To assess whether these prediction errors are significantly improved by adding a network component, we perform another hypothesis test comparing the prediction error of party-line votes versus all other votes for the full latent and network model. We observe in the second row of Table 6 that the fused

model substantially decreases MAE for both types of votes. Notably, the fused model does not produce a significantly different MAE between the two groups of votes, indicating that the network captures the systematic errors committed by the latent-only model on party-line votes.

3.2.2 Intra-party split bills

Latent factors assign each senator to a fixed position in a low-dimensional continuous space, representing collective patterns as gradations along a common set of dimensions. Though this representation often captures chamber-wide partisan patterns, localized within-party factions may not align with these dimensions. Although such patterns may be accommodated by additional latent dimensions, this representation can be substantively misleading when heterogeneity in collective behavior reflects relational structure among subsets of senators rather than high dimensionality in political preferences. The conditional independence assumption further prevents the latent model from directly representing residual relational dependence among senators.

This failure is visible in Figure 2 which displays roll call bills in the 101st Congress, positioned by Democratic and Republican yea shares, with darker shades indicating higher MAE. Latent factors are more likely to mispredict bills closer to the center, where one or both parties split internally. The 20 worst latent-predicted bills from the entire session, expressed in circled points, all feature intra-party splits close to the center point $(0.5, 0.5)$.

For further analysis, we select the set of intra-party split bills as those where at least one party has an internal minority of at least 25%. A permutation hypothesis test in the first row of Table 7 confirms that latent predictions are significantly worse for intra-party votes than all other votes. The second row performing an equivalent hypothesis test for the fused latent and network model displays a substantial reduction in MAE with the addition of a network component. However, the network component does not close the relative gap in MAE between intra-party split bills versus all others. This persisting gap indicates a possible distinction between stable within-party coalitions that recurrently vote together across multiple bills, while transient issue-specific coalitions that activate on very few bills and may therefore not generate stable pairwise relationships.

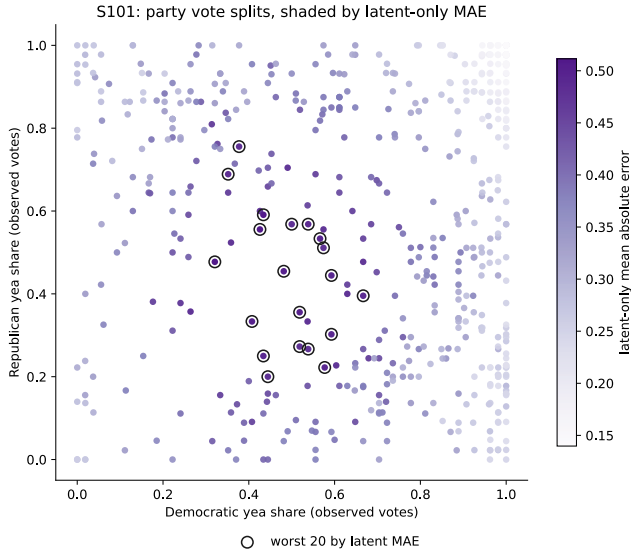


Figure 2: Roll call votes in the 101st Senate positioned by Democratic and Republican yea shares, shaded by latent-only MAE. Circled points are the 20 worst-predicted bills.

Model	Intra-party MAE	Other MAE	Gap	p
Latent only	0.364	0.215	+0.149	< 0.001
Full $\hat{L} + \hat{S}$	0.268	0.116	+0.151	< 0.001

Table 7: Prediction error on intra-party split versus all other votes in the 101st Senate.

To verify whether $\hat{S}$ captures the stable within-party coalitions across intra-party split bills, we perform an additional series of hypothesis tests. First, we find that defectors are systematically more connected to each other in $\hat{S}$ than to same-party loyalists: the defector-defector edges are stronger than defector-loyalist edges on 94.6% of intra-party split bills, with a mean edge-strength ratio of 3.34 (5000 permutations, $p < 0.001$). Second, we examine whether $\hat{S}$ captures stable within-party defection patterns across intra-party split bills by a permutation test comparing binary edge indicators of $\hat{S}$ with defection residual correlation, identified by the correlation between the differences of each senator’s vote and their latent-only predicted probability of a yea-vote. We find that within-party pairs connected in $\hat{S}$ are dramatically more correlated on split-bill residuals than non-edge pairs, with a gap of approximately 0.21 and p-value less than 0.001 for both Democrats and Republicans.

These results demonstrate that $\hat{S}$ encodes stable co-defection structures behind intra-party split bills

even when it cannot fully correct the prediction error. The remaining MAE gap possibly reflects transient coalitions that do not create recoverable patterns, while the stable coalitions captured in $\hat{S}$ are statistically distinguishable.

3.3 Identifying Collective Behavior from the Network

To examine what collective voting patterns $\hat{S}$ captures beyond the chamber-wide structure represented by the latent factors, we fit a degree-corrected stochastic block model (DC-SBM) to $\hat{S}$ alone, with no additional input such as party labels, estimated latent positions, or roll call outcomes. The DC-SBM partitions senators into K blocks based on shared patterns of co-voting relationships across $\hat{S}$. The degree correction controls for the aggregate co-voting volume of each senator, allowing block assignments to capture pattern of relationships rather than intensity. We fit the model for $K = 2, \dots, 6$ and use BIC to select $K = 5$.

Table 8 displays the five recovered blocks. Two blocks correspond to partisan cores: C1 (20 mainstream Democrats, mean extremity 0.74) and C4 (23 mainstream Republicans, mean extremity 0.52). These two groups have the largest latent positional extremity and are predicted relatively well by the latent factor model. Their presence confirms that the block model is recovering structural signals.

The three remaining blocks C0 (13 conservative Democrats, mean extremity 0.34), C2 (23 moderate Republicans, mean extremity 0.44), and C3 (20 southern Democrats, mean extremity 0.42) are substantively more interesting, occupying the lower-extremity positions in the latent space. Figure 3 displays their overlap in the latent factor space. Many conservative Democrats in C0 and southern Democrats in C3 have latent loadings overlapping with Republicans, especially moderate Republicans in C2.

While these blocks have overlapping partisan membership and latent factor loadings, they are differentiable by their patterns of cross-party co-voting on central issues. To determine the bills that serve as drivers of the blocks, we compute the *mean centered residual coproduct* over all observed pairs within the block for each roll call: $\frac{1}{|\text{pairs}|} \sum_{i < j, i, j \in \text{block}} (y_i - \hat{p}_i)(y_j - \hat{p}_j)$. This quantity is large when block members systematically vote together beyond what the latent factor predicts.

Table 9 lists the top 5 bill drivers by block, which match historical descriptions. C3 southern

Block	Label	n	Composition	Mean extremity	Mean misclass. rate
C0	Conservative Democrats	13	13D / 0R	0.34	0.260
C3	Southern Democrats	20	20D / 0R	0.42	0.193
C2	Moderate Republicans	23	1D / 22R	0.44	0.211
C4	Mainstream Republicans	23	0D / 23R	0.52	0.070
C1	Mainstream Democrats	20	20D / 0R	0.74	0.074

Table 8: DC-SBM blocks fitted to $\hat{S}$ for the 101st Senate, sorted by mean positional extremity. Mean misclassification rate is computed on the 20 worst party-line votes.

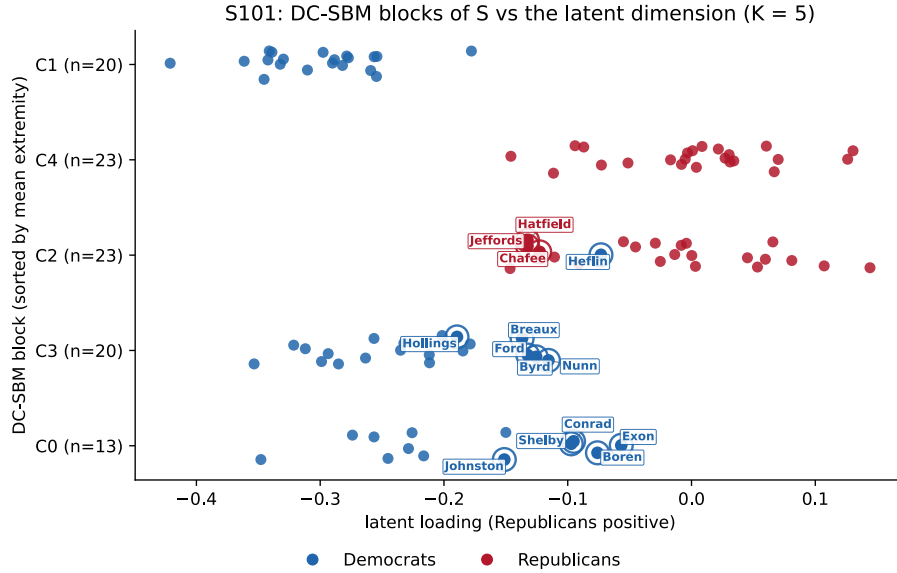


Figure 3: DC-SBM block assignments plotted against the latent factor loading for the 101st Senate, with blocks sorted by mean positional extremity. Color indicates party.

Block	Domain	Representative bills
C0	Defection from majority	Ethics Reform Act HR3660 (roll 308); budget resolution HCONRES106 (roll 64); farm bill S2830 crop insurance (roll 488)
C3	Fiscal conservatism / military	Capital gains HR3628 (rolls 295, 298); Tower nomination PN129 (roll 20); Air Force appropriations HR4404 (roll 385)
C2	Social welfare / environmental	Child-care EITC S5 (roll 99); Clean Air Act renewables S1630 (roll 358); affordable housing S566 (roll 435)

Table 9: Top bill drivers by block in the 101st Senate, ranked by mean within-block residual co-voting beyond the latent factor. Campaign finance bill S137 appears as a lower-ranked driver for both C0 and C3 and is omitted.

democrats are characterized by an alignment on fiscal conservatism or military issues on several salient bills from the early Bush administration: supporting the capital gains tax cut, the Tower defense nomination, and Air Force appropriations. C2 moderate Republicans, famously of the Northeast and Pacific Northwest, cross party lines in the opposite direction, voting with Democrats on social welfare and environmental bills: the child-care earned income tax credit, Clean Air Act renewable energy incentives, and affordable housing. C0 conservative Democrats defect from Democratic or bipartisan majorities on a broader range of issues, but with stronger interests on fiscal processes and the federal crop insurance, consistent with the boll weevil coalition of fiscally conservative Democrats from agricultural states.

These descriptions explain the placement of Heflin (D-AL) in C2 instead of C0 or C3: his factor loading is indistinguishable from other conservative Democrats in C0, but his co-voting patterns on social and environmental votes align him with the moderate Republican block, a distinction the latent factor misses and the network recovers.

Findings from DC-SBM again highlight the failure of latent factor models in predicting moderate-positional senators. The last column of Table 8 shows that the three relatively moderate blocks, C0, C2, and C3, have substantially high mean misclassification rates on the 20 worst latent-predicted bills compared to the other two mainstream groups. The most misclassified senators in each group occupy centrist, overlapping latent positions in Figure ??: Conrad, Exon, and Shelby in C0 (65–70% misclassification rate); Nunn and Breaux in C3 (75–79%); and Chafee and Hatfield in C2 (71–84%). While the fitted latent factors do not distinguish between these senators, $\hat{S}$ captures relational structure that is obscured in the latent representation, and the DC-SBM makes that structure interpretable.

3.4 Changes Over Time

Many congressional scholars recognize the polarizing partisan and ideological shifts that have occurred in both chambers of the Congress over the past few decades. These changes also appear on latent factors: Caughey and Schickler (2016) argue that the two-dimensional latent factors of NOMINATE do not maintain consistent ideological meaning over time. In this section, we attempt to characterize these changes through both the latent factor and network components of the model. Our previous analyses

focused on the 101st Senate (1989-90), and we additionally examine the 114th Senate (2015-16) for comparison.

3.4.1 Latent Factors

The increased partisan dominance due to polarization creates two especially notable changes in the latent factors of the 114th Senate, compared to S101. First, latent factors mostly reduce to one meaningful dimension that captures strong partisan signals. Table 10 compares the correlation between fitted ideal points of bills versus the proportion of Democrat and Republican yea votes per bill across the 101st and 114th Senates. While the diagonal correlations are extremely high for both sessions, the off-diagonals are substantially larger for the 114th Senate, indicating that the second factor carries a sizable partisan signal that is already captured by the first factor. This change is consistent with partisan polarization reducing Senate voting onto a single partisan dimension.

	S101		S114	
	p_D	p_R	p_D	p_R
$\hat{\theta}_{\text{var},1}$	-0.984	-0.047	-0.990	+0.516
$\hat{\theta}_{\text{var},2}$	+0.128	+0.988	+0.434	-0.986

Table 10: Correlation between the estimated latent bill factors ($\hat{\theta}_{\text{var},1}, \hat{\theta}_{\text{var},2}$) and the proportion of Democratic and Republican yea votes per bill (p_D, p_R) for the 101st and 114th Senates.

Model	Party-line MAE	Other MAE	Gap	p
Latent only	0.221	0.208	+0.013	0.092
Full $\hat{L} + \hat{S}$	0.136	0.142	-0.005	0.418

Table 11: Prediction error for positionally moderate senators on party-line versus all other votes, 114th Senate.

Second, with polarization decreasing the number of moderates, the systematic failure of latent models on moderates’ party-line votes described in section 3.2.1 accounts for fewer mispredictions. While the list of worst latent-predicted votes ordered by MAE was predominated by party-line bills in the 101st senate, intra-party split bills dominate an equivalent list for the 114th Senate. We examine whether the latent

model is systematically worse at predicting party-line bills in the 114th Senate through hypothesis tests listed in Table 11, which replicates Table 6 for the 114th Senate. The latent-only gap between party-line and all other bills is still positive but significant only at the 0.1 level, indicating that the systematic pattern still exists but occurs less. The null result for the second test indicates that the network still systematically captures these mispredictions and reduce errors from the latent-only model.

3.4.2 Shifted Blocks in the Network

To analyze the changed block structure of $\hat{S}$ for the 114th senate, we repeat the DC-SBM analysis, where BIC selects $K = 5$ blocks. Table 12 displays the five recovered blocks. Just like the 101st senate, we have two mainstream groups: C0 (24 members, mainstream Democrats, mean extremity 0.608) and C1 (19 members, mainstream Republicans, mean extremity 0.631).

The three remaining blocks have changed markedly from the 101st senate. Rather than issue-driven “ideological” wings within the party, these blocks reflect institutional and procedural positioning within a polarized chamber. C3 (21 members, governing bloc, mean extremity 0.265) has the lowest extremity and is cross-party, combining Republican floor leadership (McConnell, Cornyn, Hatch, Alexander) and appropriators (Cochran, Wicker), moderate Republicans (Collins, Kirk, Murkowski, Ayotte, Portman), and four Democrats (Boxer, Durbin, Peters, Stabenow). C2 (22 members, cross-party moderates, mean extremity 0.400) is also cross party, with a handful of red-state Democrats (Manchin, Heitkamp, Tester, McCaskill). C4 (14 members, insurgent bloc, mean extremity 0.646) contains the insurgent Republican wing (Lee, Cruz, Sessions, Cotton, Sasse) who consistently vote against leadership-negotiated deals and procedural motions. The placement of two Democrats (Feinstein, Shaheen) in this group likely reflects specific cross-cutting votes rather than an issue-based alignment with the rest of the block.

From Figure 4 displaying the range of latent loadings by block for the 114th senate, we see a cleaner party-line split in the latent space compared to S101, speaking to the level of polarization in the congress. The most misclassified senators are still relative centrists; however, the larger divide between the parties and the lack of overlap points to a lack of genuine positional moderates. The latent composition of each block also provides more contrast. C2 and C3 are cross-party blocks spanning a wide range of the latent

Block	Label	n	Composition	Mean extremity	Mean misclass. rate
C3	Governing bloc	21	4D / 17R	0.265	0.136
C2	Cross-party moderates	22	15D / 6R	0.400	0.109
C0	Mainstream Democrats	24	23D / 1I / 0R	0.608	0.004
C1	Mainstream Republicans	19	0D / 19R	0.631	0.064
C4	Insurgent bloc	14	2D / 12R	0.646	0.091

Table 12: DC-SBM blocks fitted to $\hat{S}$ for the 114th Senate, sorted by mean positional extremity. Mean misclassification rate is computed on the 20 worst party-line votes.

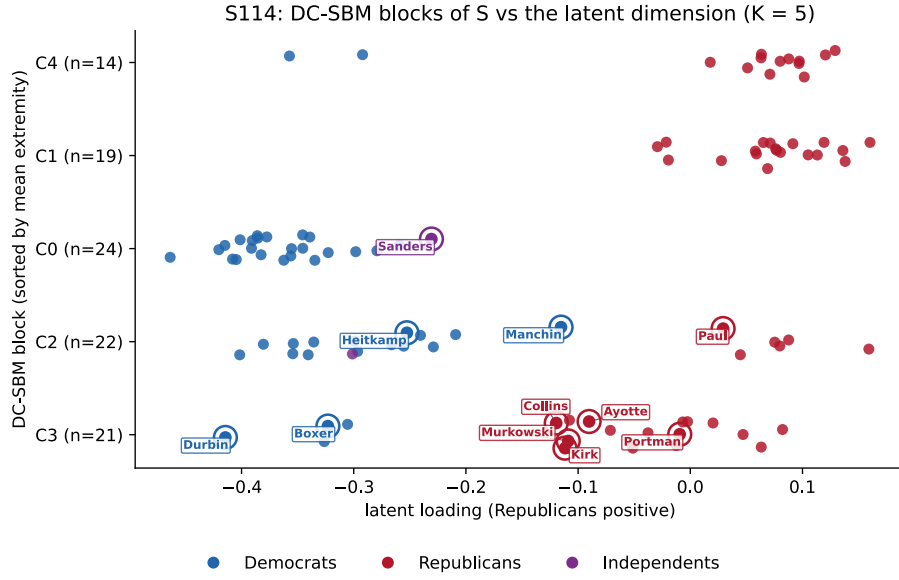


Figure 4: DC-SBM block assignments plotted against the latent factor loading for the 114th Senate, with blocks sorted by mean positional extremity. Color indicates party.

Block	Domain	Representative bills
C3	Anti-insurgent governance	ESEA parental opt-out S1177 (roll 235); regional commissions HR2028 (roll 395); military construction HR2029 (roll 337)
C2	Cross-party privacy / energy	CISA cyber threat review S754 (roll 289); Keystone pipeline delay S1 (rolls 31, 47); terrorism risk insurance HR26 (roll 1)
C4	Anti-establishment defection	Legislative branch approps HR5325 (rolls 485, 486); Zika response HR2577 (roll 415); firearms transfer delay HR2578 (roll 449)

Table 13: Top bill drivers by block in the 114th Senate, ranked by mean within-block residual co-voting beyond the latent factor. Mainstream Democrat (C0) and mainstream Republican (C1) blocks are omitted; their drivers reflect standard party-line voting captured by the latent factor.

space, whereas none of the blocks in S101 were cross-party.

Examining top bill drivers by blocks in Table 13 suggests that these blocks may signal institutional, rather than policy, divides. The top drivers of C3 do not intersect in their issue content, but they are all amendments proposed by insurgent Republicans in C4 challenging leadership-negotiated deals on education, regional spending, and military construction. In other terms, this block reflects an institutional cross-party effort for legislative collaboration, rather than shared issue interests. C4’s top drivers mirror this dynamic, featuring bills containing the Republican leadership’s negotiated fiscal deals that the block largely voted against. In this context, Feinstein and Shaheen’s placement in C4 is likely a result of a limited K , the number of blocks, because they have near-zero edge weights with other members of the group; C4 was still likely a better fit for them because they are even less connected with other Democrats. A larger K may have placed them in a separate block as atypical Democrats.

C2, on the other hand, is the only block formed by cross-party policy agreement. Its drivers are CISA privacy amendments and Keystone pipeline votes, where red-state Democrats and libertarian-leaning Republicans including Rand Paul found common ground on surveillance and energy policy, in opposition with both party leaderships. This block is the only one in S114 which resembles the issue-driven blocks of S101, representing a small residual of policy-based cross-party collaboration in an institutionally divided chamber.

4 Discussion

Legislative behavior is inherently collective, yet latent factors, which have become the methodological linchpin of roll call analysis, have inherent barriers to representing group behavior. While they can represent collective behavior as chamber-wide gradations along common dimensions, coalitional patterns can be obscured in the latent space, and the conditional independence assumption categorically sets aside relational effects.

We have demonstrated that the fused latent factor and network model is useful not only in improving upon latent predictions but also in analyzing collective patterns of roll call behavior that remain after

accounting for chamber-wide latent patterns, such as geographic proximity of senators’ constituencies, centrist voting blocs on party-line bills, and within- and across-party factions. Compared to existing network approaches that attribute all signal to relational structure, our approach is capable of helping analyze coalitional patterns on a network while capturing chamber-wide patterns with latent factors. Comparing S101 and S114, the network gives a much more detailed insight into the deeper changes in the motivations underlying collective roll call behavior, from policy positions and possibly ideology to partisanship and institutional obstruction. This change is not apparent from studying only latent models.

The opposition between governing versus insurgent cross-party blocks in S114 was recovered from roll call data alone using our fused model and DC-SBM without any additional input on party or issue interests. This suggests that network-based methods can be used to detect coalitions and voting blocs, even in legislatures where these groupings are not formally defined. While this paper is partially focused on providing some “supervised” validation of the model utilizing a well-studied dataset and existing historical analysis, for future work, this model could be instrumental in “unsupervised” discoveries and measurements of collective roll call behavior within both federal and local legislatures.

The analysis of intra-party split bills suggests a distinction between stable coalitions that were captured in $\hat{S}$ and transient coalitions that may not generate recoverable relational patterns. This distinction illustrates that some collective legislative behavior is structural and recoverable from behavioral data, while other patterns may be situational and temporary. This application could supply debates in the political literature about whether congressional coalitions are durable interest-based groups or ad-hoc strategic voting.

Several limitations of our network analysis must be acknowledged. First, $\hat{S}$ has reduced intra-party resolution with higher polarization. In S114, cross-party co-voting is rare, leading to very low edge weights for most cross-party senator pairs. This impedes the DC-SBM’s capability to capture within-party coalitions. Second, the interpretation of voting blocs is heavily dependent on the choice of K . While the BIC can help select a natural fit, a larger K may offer more exact fits, for instance for Feinstein and Shaheen in S114 who were misplaced in the insurgent bloc. However, a larger K may obscure

interpretation. Third, relatedly, our approach currently lacks analysis of exactly what signals get captured as stable rather than disregarded as transient.

While our model is not identical to NOMINATE, latent factor models broadly share the foundational mechanisms that produce the systematic failures identified in this paper: the conditional independence assumption and the representation of collective behavior through chamber-wide dimensions. Therefore, the intuition behind this critique transfers beyond our specific application.

Our analysis broadly suggests that roll call data alone, without additional institutional or relational data, contains recoverable traces of collective legislative behavior, but latent factor models alone can obscure their relational patterns. As coalitions become less identifiable from policy directions alone, network models that can directly encode relational structures may be increasingly helpful for understanding the collective nature of roll call votes.

Acknowledgments

For helpful comments, we thank Zhiliang Yung, Yunxiao Chen, Naoki Egami, Michael Sobel, Jacob Montgomery, Yiqing Xu, Shahryar Minhas, poster session participants at the 40th Annual Meeting of the Society for Political Methodology, and panel participants at the 2026 APSA Annual Meeting.

References

- Abramowitz, Alan I. (2012). “US Senate Elections in a Polarized Era”. Vol. 27. CQ Press, p. 30.
- Arnold, Laura W., Rebecca E. Deen, and Samuel C. Patterson (2000). “Friendship and votes: The impact of interpersonal ties on legislative decision making”. *State and Local Government Review* 32(2), pp. 142–147.
- Battaglini, Marco, Valerio Leone Sciabolazza, and Eleonora Patacchini (2020). “Effectiveness of Connected Legislators”. *American Journal of Political Science* 64(4), pp. 739–756.

- Boyd, Stephen, Neal Parikh, Eric Chu, Borja Peleato, and Jonathan Eckstein (2011). “Distributed optimization and statistical learning via the alternating direction method of multipliers”. *Foundations and Trends in Machine learning* 3(1), pp. 1–122.
- Caughey, Devin and Eric Schickler (2016). “Substance and change in congressional ideology: NOMINATE and its alternatives”. *Studies in American Political Development* 30(2), pp. 128–146.
- Chen, Yunxiao, Xiaou Li, Jingchen Liu, and Zhiliang Ying (2016). “A fused latent and graphical model for multivariate binary data”. *arXiv preprint arXiv:1606.08925*.
- Clinton, Joshua, Simon Jackman, and Douglas Rivers (2004). “The statistical analysis of roll call data”. *American Political Science Review* 98(2), pp. 355–370.
- Fafchamps, Marcel and Flore Gubert (2007). “The Formation of Risk Sharing Networks”. *Journal of Development Economics* 83(2), pp. 326–350.
- Fowler, James H. (2006). “Connecting the Congress: A study of cosponsorship networks”. *Political analysis* 14(4), pp. 456–487.
- Heckman, James J. and James M. Snyder (1997). “Linear Probability Models of the Demand for Attributes with an Empirical Application to Estimating the Preferences of Legislators”. *The RAND Journal of Economics* 28, S142–S189.
- Kim, Jang Hyun, George A. Barnett, and K. Hazel Kwon (2010). “The influence of social networks on the US Senate roll-call voting”. *International Journal of E-Politics (IJEP)* 1(4), pp. 22–41.
- Kirkland, Justin H (2011). “The relational determinants of legislative outcomes: Strong and weak ties between legislators”. *The Journal of Politics* 73(3), pp. 887–898.
- Kirkland, Justin H. and Justin H. Gross (2014). “Measurement and theory in legislative networks: The evolving topology of Congressional collaboration”. *Social Networks* 36. Special Issue on Political Networks, pp. 97–109.
- Poole, Keith T. and Howard Rosenthal (1985). “A Spatial Model for Legislative Roll Call Analysis”. *American Journal of Political Science* 29(2), pp. 357–384.

- Porter, Mason A., Peter J. Mucha, Mark E.J. Newman, and Casey M. Warmbrand (2005). “A network analysis of committees in the US House of Representatives”. *Proceedings of the National Academy of Sciences* 102(20), pp. 7057–7062.
- Sinclair, Barbara (2016). *Unorthodox lawmaking: New legislative processes in the US Congress*. CQ Press.
- Stauffer, Dietrich (2008). “Social applications of two-dimensional Ising models”. *American Journal of Physics* 76(4), pp. 470–473.
- Wawro, Gregory J. and Eric Schickler (2007). *Filibuster: Obstruction and lawmaking in the US Senate*. Vol. 134. Princeton University Press.
- Wright, John and Yi Ma (2022). *High-dimensional data analysis with low-dimensional models: Principles, computation, and applications*. Cambridge University Press.